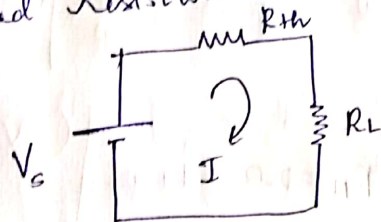


THIRD SEMESTER BTECH DEGREE EXAM, DEC 2020

(2019 SCHEME)

EET201 CIRCUITS AND NETWORKS

- ① Maximum power transfer theorem states that maximum power is delivered to the load by the source only when equivalent resistance of the source becomes equal to the load resistance.



$$I = \frac{V_s}{R_s + R_L}$$

Power at load $P = I^2 R_L$

$$P = \left(\frac{V_s}{R_s + R_L} \right)^2 R_L$$

For maximum power $P_{max} = \frac{dP}{dR_L} = 0$

$$\frac{d}{dR_L} \left(\frac{V_s^2 R_L}{(R_s + R_L)^2} \right) = 0$$

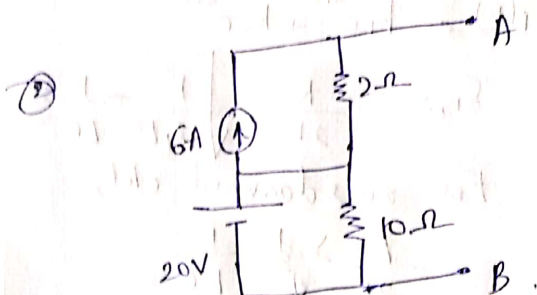
$$V_s^2 \left[\frac{(R_s + R_L)^2 - 2R_L(R_s + R_L)}{(R_s + R_L)^4} \right] = 0$$

$$(R_s + R_L)^2 = 2R_L(R_s + R_L)$$

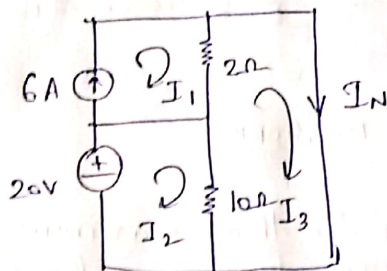
$$R_s + R_L = 2R_L$$

$$\boxed{R_s = R_L}$$

Hence proved.



Short circuiting A & B to find I_N



$$I_1 = 6A \rightarrow (1)$$

$$20 - 10(I_2 - I_3) = 0$$

$$-10I_2 + 10I_3 = -20 \rightarrow (2)$$

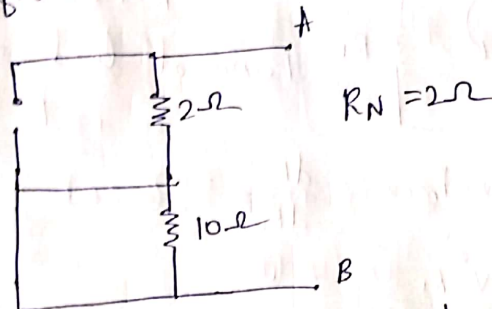
$$-10(I_3 - I_2) - 2(I_3 - I_1) = 0$$

$$-10I_3 + 10I_2 - 2I_3 + 12 = 0$$

$$10I_2 - 12I_3 = -12 \rightarrow (3)$$

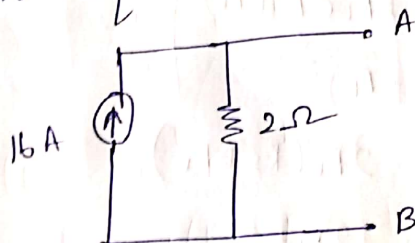
$$I_N = I_3 = 16A$$

To find R_N



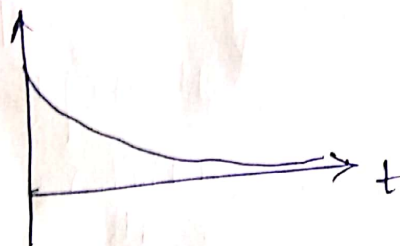
$$R_N = 2\Omega$$

Norton Equivalent circuit



② Based on damping ratio, there are 4 types of RLC circuits.

① Overdamped ($\zeta > 1$)



$$i(t) = k_1 e^{s_1 t} + k_2 e^{s_2 t}$$

$$s_{1,2} = \frac{-R}{2L} \pm \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}$$

For overdamped ckt,

$$\frac{R}{2L} > \frac{1}{LC}$$

(b) Critically damped ($\xi = 1$)

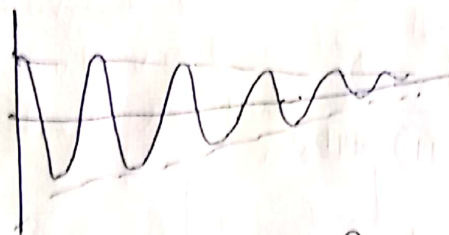


Here $\frac{R}{2L} = \frac{1}{LC}$

$$i(t) = e^{-\alpha t} (k_1 + k_2 t)$$

g.

(c) Underdamped ($0 < \xi < 1$)

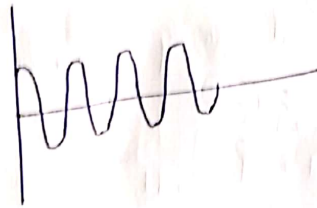


Here $\frac{R}{2L} < \frac{1}{LC}$

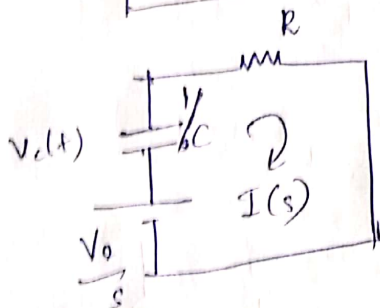
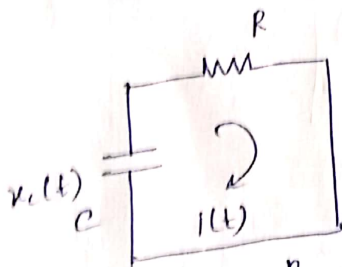
$$i(t) = e^{-\alpha t} [k_1 \cos \beta t + k_2 \sin \beta t]$$

$$s_{1,2} = \alpha \pm j\beta$$

(d) undamped ($\xi = 0$)



(4)



given $v_c(0^-) = v_c(0^+) = V_0$

$$\frac{V_0}{s} - \frac{1}{sC} I(s) - R I(s) = 0$$

$$\frac{V_0}{s} = I(s) \left[R + \frac{1}{sC} \right]$$

$$\frac{V_0}{s} = I(s) \left[\frac{sRC + 1}{sC} \right]$$

$$I(s) = \frac{V_0 C}{sRC + 1}$$

$$I(s) = \frac{V_0 / R}{s + \frac{1}{RC}}$$

$$i(t) = \frac{V_0}{R} e^{-t/RC}$$

$$\begin{aligned}
 v_c(t) &= \frac{1}{C} \int i(t) dt \\
 &= \frac{1}{C} \int \frac{V_0}{R} e^{-t/RC} dt \\
 &= \frac{-V_0}{RC} \frac{e^{-t/RC}}{-1/RC}
 \end{aligned}$$

$$v_c(t) = V_0 e^{-t/RC}$$

⑤
$$I(s) = \frac{2s+4}{s^2+4s+3} = \frac{2(s+2)}{(s+1)(s+3)}$$

$$I(s) = \frac{A}{s+1} + \frac{B}{s+3}$$

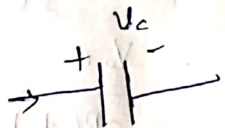
$$A = I(s)(s+1) \Big|_{s=-1} = \frac{2}{2} = 1$$

$$B = I(s)(s+3) \Big|_{s=-3} = \frac{2(-1)}{-2} = 1$$

$$I(s) = \frac{1}{s+1} + \frac{1}{s+3}$$

$$i(t) = e^{-t} + e^{-3t}$$

$$\begin{aligned}
 v(t) &= i(t) \times R \\
 &= \underline{\underline{2(e^{-t} + e^{-3t})}}
 \end{aligned}$$



$$v_c = \frac{1}{C} \int i(t) dt$$

$$i(t) = C \frac{dv_c(t)}{dt}$$

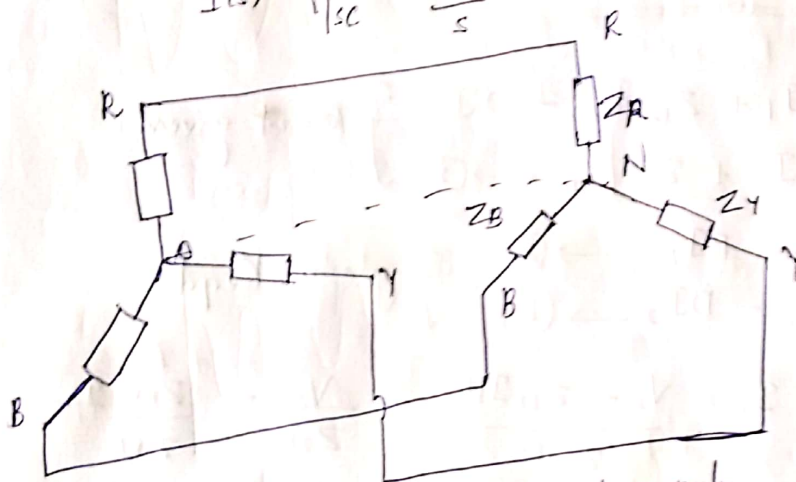
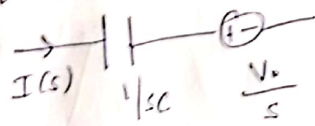
Applying Laplace transform

$$I(s) = C[sV(s) - v(s)]$$

$$= C[sV(s) - V_0]$$

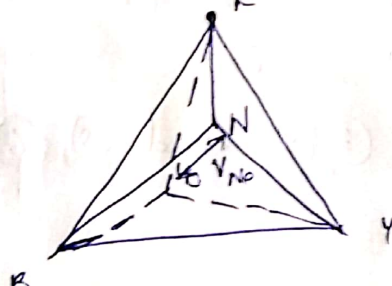
$$V(s) = \frac{V_0}{s} + \frac{I(s)}{sC}$$

Equivalent circuit



O $\rightarrow$ neutral point of supply
N $\rightarrow$ neutral point of load.

For 3 phase 3 wire unbalanced system, the difference b/w the potential at neutral point of supply and load is not zero. This shift in potential is neutral shift. In 4 wire system, a current I_N through ON line is given by $I_N = I_R + I_Y + I_B$.



⑧ Q factor is defined as the ratio of voltage across the inductor or capacitor to the resonant voltage for a series RLC circuit

$$Q = \frac{V_{L0}}{V_0} = \frac{V_{C0}}{V_0}$$

$$Q = \frac{I_0 X_L}{I_0 R} = \frac{L \omega_0 I_0}{R I_0} = \frac{L}{R} \frac{1}{\sqrt{LC}}$$

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

⑨ $V_1 = Z_{11} I_1 + Z_{12} I_2 \rightarrow (1)$ } Reqd given

$V_2 = Z_{21} I_1 + Z_{22} I_2 \rightarrow (2)$

$V_1 = A V_2 - B I_2 \rightarrow (3)$ } given, Reqd

$I_1 = C V_2 - D I_2 \rightarrow (4)$

From (4) $I_1 = \frac{V_2 - Z_{22} I_2}{Z_{21}} = \frac{V_2}{Z_{21}} - \frac{Z_{22}}{Z_{21}} I_2$

$$V_1 = Z_{11} \left(\frac{V_2 - Z_{22} I_2}{Z_{21}} \right) + Z_{12} I_2$$

$$V_1 = \frac{Z_{11}}{Z_{21}} V_2 + \left(\frac{Z_{12} Z_{21} - Z_{11} Z_{22}}{Z_{21}} \right) I_2$$

$$V_1 = \frac{Z_{11}}{Z_{21}} V_2 - \left(\frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{21}} \right) I_2 \rightarrow (5)$$

$$I_1 = \frac{V_2}{Z_{21}} - \frac{Z_{22}}{Z_{21}} I_2 \rightarrow (6)$$

Comparing eq^s (3), (4) and (5), (6) we get

$$12 - 10I_1 + 2(I_1 - I_2) - 4(I_1 - I_2) = 0$$

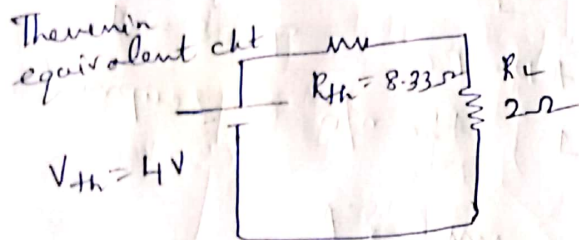
$$-12I_1 + 2I_2 = -12 \rightarrow (1)$$

$$-4(I_1 - I_2) - 5I_2 = 0$$

$$4I_1 - 9I_2 = 0 \rightarrow (2)$$

$$I_{sc} = I_2 = 0.48A$$

$$R_{th} = \frac{V_{th}}{I_{sc}} = \frac{4}{0.48} = 8.33\Omega$$

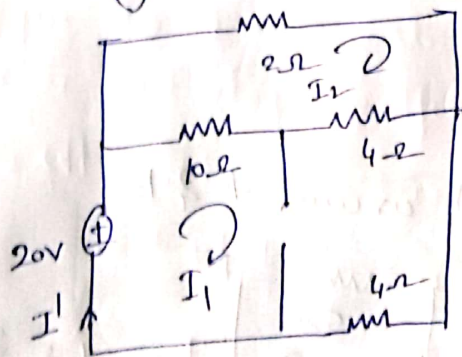


Power dissipated in $2\Omega = I^2 R_L$

$$I = \frac{4}{2 + 8.33} = 0.387A$$

$$P = I^2 R_L = 0.299W$$

12a) Considering 20V source alone



$$20 - 14(I_1 - I_2) - 4I_1 = 0$$

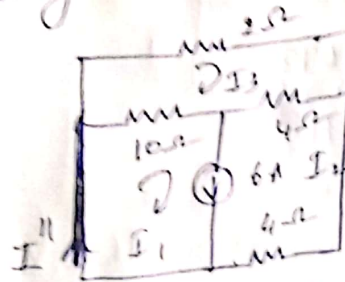
$$-18I_1 + 14I_2 = -20 \rightarrow (1)$$

$$-2I_2 - 14(I_2 - I_1) = 0$$

$$14I_1 - 16I_2 = 0$$

$$I' = I_1 = 3.478A$$

considering 6A source alone



$$-10(I_1 - I_2) - 4(I_2 - I_3) - 4I_2 = 0$$

$$-10I_1 - 8I_2 + 4I_3 = 0$$

$$I_1 - I_2 = 6$$

$$-2I_3 - 4(I_3 - I_2) - 10(I_3 - I_1) = 0$$

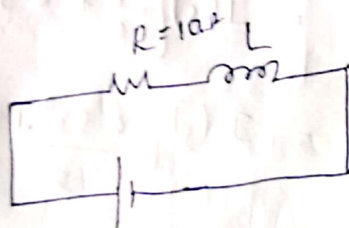
$$10I_1 + 4I_2 - 16I_3 = 0$$

$$I'' = I_1 = 4.69 \text{ A}$$

$$I = I' + I'' = 4.69 + 3.48 = 8.17 \text{ A}$$

power dissipated by 20V source $= V \times I$
 $= 20 \times 8.17$
 $= 163.4 \text{ W}$

13 a



$$i = I_0 (1 - e^{-t/\tau})$$

$$\tau = \frac{L}{R} = \frac{L}{10}$$

$$I_0 = \frac{50}{10} = 5 \text{ A}$$

$$i = 2.5, t = 1 \text{ s}$$

$$2.5 = 5 (1 - e^{-10/L})$$

$$0.5 = 1 - e^{-10/L}$$

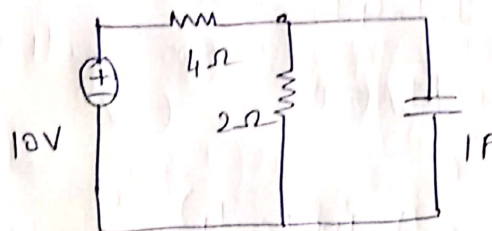
$$-10/L$$

$$e = 0.5$$

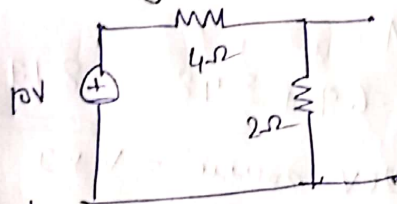
$$\frac{10}{L} = 0.693$$

$$L = 14.4 \text{ H}$$

At $t = 0^-$

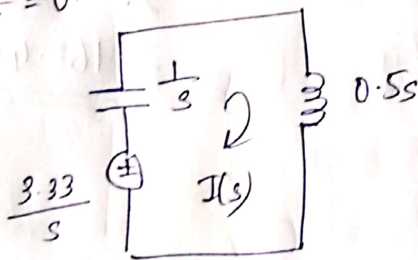


At steady state, capacitor will act as open circuit



$$V_c(0^-) = V_c(0^+) = V_c(\infty) = \frac{10 \times 2}{4+2} = 3.3$$

At $t = 0^+$



$$\frac{3.33}{s} - \frac{I(s)}{s} - 0.5sI(s) = 0$$

$$\frac{3.33}{s} = I(s) \left[\frac{1}{s} + 0.5s \right]$$

$$\frac{3.33}{s} = I(s) \left[\frac{0.5s^2 + 1}{s} \right]$$

$$I(s) = \frac{3.33}{0.5s^2 + 1}$$

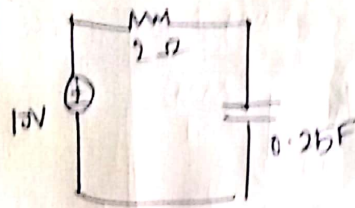
$$= \frac{6.66}{s^2 + 2}$$

$$I(s) = \frac{6.66}{\sqrt{2}} \times \frac{\sqrt{2}}{s^2 + (\sqrt{2})^2}$$

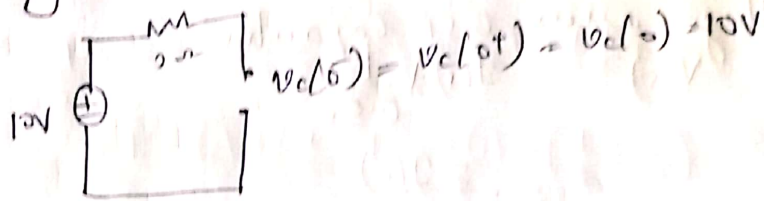
$$i(t) = 4.71 \sin \sqrt{2} t$$

(11a)

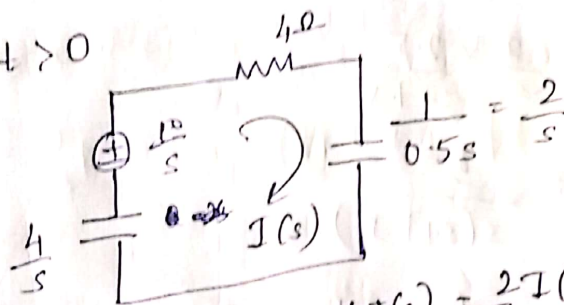
$$At t = 0^-$$



At steady state, capacitor will act as open circuit



$$At t > 0$$



$$\frac{10}{s} - \frac{4}{s}I(s) - 4I(s) - \frac{2}{s}I(s) = 0$$

$$\frac{10}{s} - 4I(s) - \frac{6}{s}I(s) = 0$$

$$\frac{10}{s} = I(s) \left[4 + \frac{6}{s} \right]$$

$$\frac{10}{s} = I(s) \left[\frac{4s + 6}{s} \right]$$

$$I(s) = \frac{10}{4s + 6}$$

$$= \frac{10/4}{s + 6/4}$$

$$I(s) = \frac{2.5}{s + 1.5}$$

$$V_C(t) = \frac{1}{C} \int i(t) dt$$

$$v_c(t) = \frac{5}{0.5} \int \frac{1}{s+1.5} dt$$

$$= \frac{5}{0.5} \int \frac{1}{s+1.5} dt$$

$$v_c(t) = \frac{5}{0.5} \frac{e^{-1.5t}}{-1.5}$$

Voltage across 0.5F Capacitor

$$V(s) = I(s) \times \frac{1}{0.5s}$$

$$= \frac{2I(s)}{s}$$

$$= \frac{2 \times 2.5}{s} \frac{1}{s+1.5}$$

$$= \frac{5}{s(s+1.5)}$$

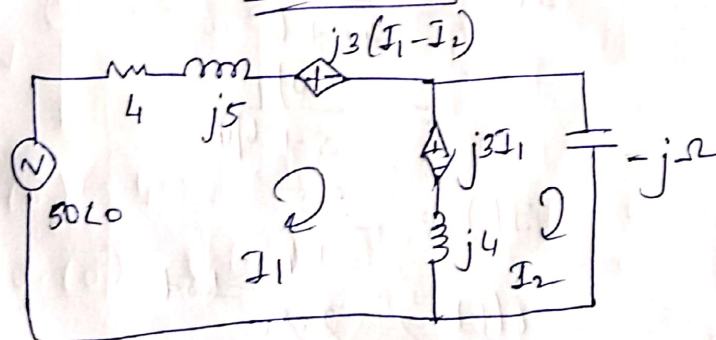
$$= \frac{A}{s} + \frac{B}{s+1.5}$$

$$A = 3.33$$

$$B = -3.33$$

$$V(s) = \frac{3.33}{s} - \frac{3.33}{s+1.5}$$

$$v_c(t) = 3.33(1 - e^{-1.5t})$$



$$50\angle 0 - (4+j5)I_1 - j3(I_1 - I_2) - j3I_1 - j4(I_1 - I_2) = 0$$

$$(-4-j15)I_1 + j7I_2 = -50\angle 0$$

$$-(4+j15)I_1 + j7I_2 = -50\angle 0$$

$$(4+j15)I_1 - j7I_2 = 50\angle 0 \rightarrow (1)$$

$$-j4(I_2 - I_1) + j3I_1 + jI_2 = 0$$

$$j7I_1 - j3I_2 = 0 \rightarrow (2)$$

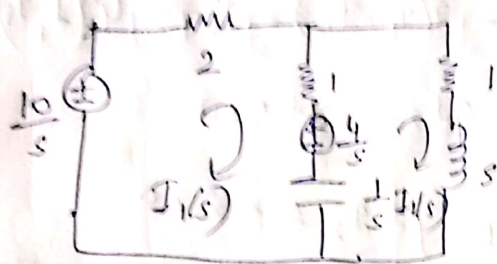
$$\begin{bmatrix} 4+j15 & -j7 \\ j7 & -j3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 50\angle 0 \\ 0 \end{bmatrix}$$

$$I_2 = \frac{\Delta_2}{\Delta}$$

$$= \frac{\begin{vmatrix} 4+j15 & 50\angle 0 \\ j7 & 0 \end{vmatrix}}{\begin{vmatrix} 4+j15 & -j7 \\ j7 & -j3 \end{vmatrix}}$$

$$= \frac{-j7 \times 50\angle 0}{(4+j15)(-j3) + (j7)^2}$$

$$= \underline{\underline{27.6 \angle 18.43^\circ \text{ A}}}$$



$$\frac{10}{s} - 2I_1(s) - (I_1(s) - I_2(s)) - \frac{4}{s} - \frac{1}{s}(I_1(s) - I_2(s)) = 0$$

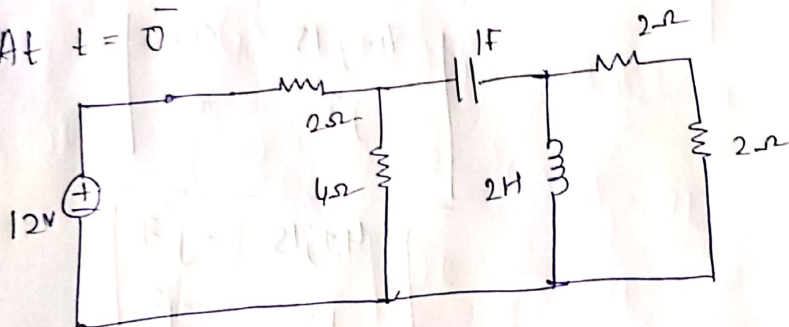
$$-(3 + \frac{1}{s})I_1(s) + (1 + \frac{1}{s})I_2(s) = -\frac{10}{s}$$

$$(s + \frac{1}{s})I_1(s) - (1 + \frac{1}{s})I_2(s) = \frac{10}{s} \rightarrow (1)$$

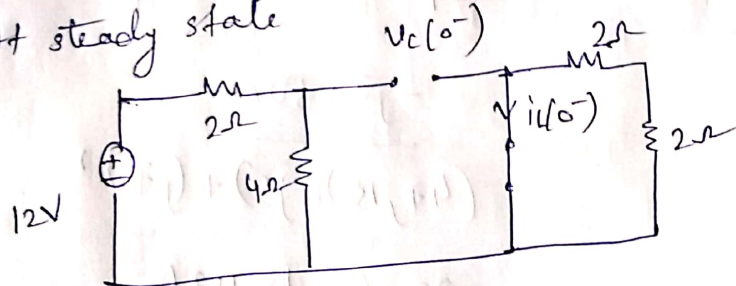
$$-(s+1)I_2(s) - (1 + \frac{1}{s})(I_2(s) - I_1(s)) + \frac{4}{s} = 0$$

$$-(1 + \frac{1}{s})I_1(s) + (s + \frac{1}{s} + 2)I_2(s) = \frac{4}{s} \rightarrow (2)$$

At $t = 0^-$



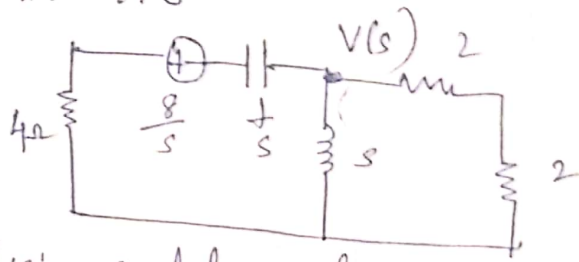
At steady state



$$i_L(0^-) = i_L(0^+) = i_L(0) = 0 \text{ A}$$

$$V_C(0^-) = V_C(0^+) = V_C(0) = 12 \times \frac{4}{6} = 8 \text{ V}$$

At $t > 0$



Using nodal analysis

$$\frac{V(s) + \frac{8}{s}}{4 + \frac{1}{s}} + \frac{V(s)}{s} + \frac{V(s)}{4} = 0$$

$$V(s) \left[\frac{1}{4 + \frac{1}{s}} + \frac{1}{s} + \frac{1}{4} \right] = \frac{-\frac{8}{s}}{4 + \frac{1}{s}}$$

$$V(s) \left[\frac{s}{4s+1} + \frac{1}{s} + \frac{1}{4} \right] = \frac{-8}{4s+1}$$

$$V(s) \left[\frac{4s^2 + 4(4s+1) + s(4s+1)}{4s(4s+1)} \right] = \frac{-8}{4s+1}$$

$$V(s) \left[\frac{4s^2 + 16 + 4 + 4s^2 + s}{4s} \right] = -8$$

$$V(s) \left[\frac{8s^2 + 17s + 4}{4s} \right] = -8$$

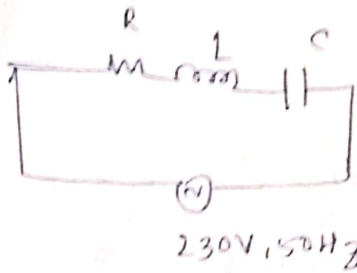
$$V(s) = \frac{-32s}{8s^2 + 17s + 4}$$

$$= \frac{-4s}{s^2 + \frac{17}{8}s + 0.5}$$

$$= \frac{-4s}{s^2 + 2 \times \frac{17}{16}s + \left(\frac{17}{16}\right)^2 - \left(\frac{17}{16}\right)^2 + 0.5}$$

$$= \frac{-4s}{(s + 1.0625)^2 - 0.5625}$$

(17)



$$I_{max} = 2A$$

$$V_C = 500V$$

(i) At resonance, $V_L = V_C = 500V$

$$I_{max} = \frac{V}{R} = \frac{230}{R} = 2$$

$$R = 115\Omega$$

$$V_L = I X_L = \omega L I$$

$$500 = 2 \times 2\pi \times 50 \times V_L$$

$$L = 0.79H$$

$$V_C = \frac{I}{\omega C} = \frac{I}{2\pi f C} = 500$$

$$C = \frac{1}{500 \times \pi \times 50} = 12.7\mu F$$

(ii)

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{115} \sqrt{\frac{0.79}{12.7 \times 10^{-6}}} = 2.168$$

$$Q = \frac{f_r}{BW}$$

$$BW = \frac{f_r}{Q} = \frac{50}{2.168} = 23Hz$$

(iii)

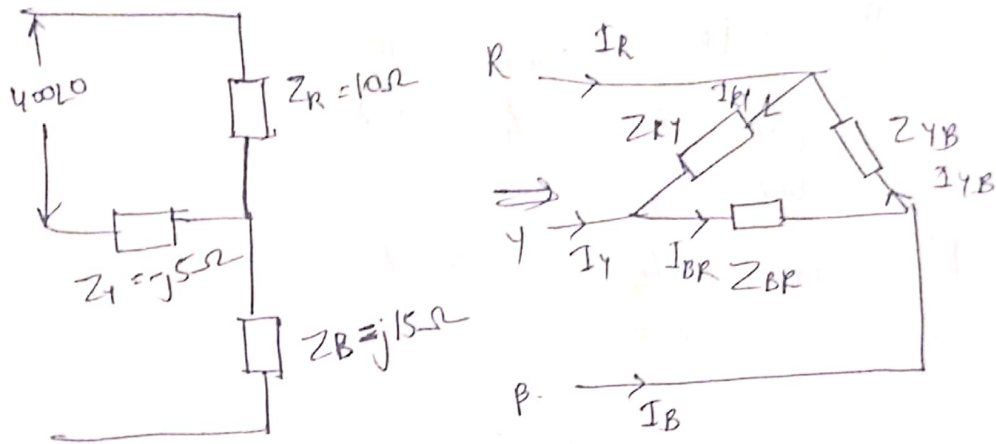
$$f_{1,2} = f_r \pm \frac{R}{4\pi L}$$

$$= 50 \pm \frac{115}{4\pi \times 0.79} = 38.42Hz$$

$$f_2 = f_r + \frac{R}{4\pi L}$$

$$= 50 + \frac{115}{4\pi \times 0.79} = 61.58Hz$$

18



$$Z_{R1} = \frac{10(j5) + 75 + j150}{j15}$$

$$= 6.66 - 5j$$

$$Z_{YB} = \frac{-50j + 75 + 150j}{10}$$

$$= 7.5 + 10j$$

$$Z_{BR} = \frac{-50j + 75 + 150j}{-j5} = -20 + 15j$$

$$V_{R1} = 400\angle 0^\circ, V_{YB} = 400\angle -120^\circ, V_{BR} = 400\angle -240^\circ$$

$$I_{R1} = \frac{400\angle 0^\circ}{6.66 - 5j} = 48.03\angle 36.87^\circ$$

$$I_{YB} = \frac{400\angle -120^\circ}{7.5 + 10j} = 32\angle -173.13^\circ$$

$$I_{BR} = \frac{400\angle -240^\circ}{-20 + 15j} = 16\angle -23.1^\circ$$

$$I_R = I_{R1} - I_{YB} = 77.4\angle 24.95^\circ$$

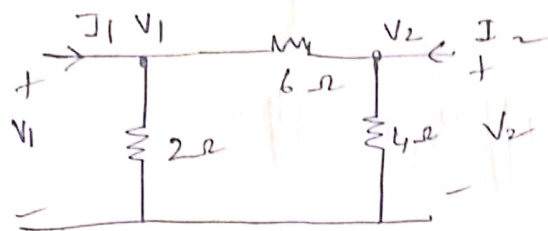
$$I_Y = I_{BR} - I_{RY}$$

$$= 42.35 \angle -124.01^\circ$$

$$I_B = I_{YB} - I_{BR}$$

$$= 46.55 \angle 176.78^\circ$$

(19)



$$-I_1 + \frac{V_1}{2} + \frac{V_1 - V_2}{6} = 0$$

$$V_1 \left(\frac{1}{2} + \frac{1}{6} \right) - \frac{V_2}{6} = I_1$$

$$\frac{2V_1}{3} - \frac{V_2}{6} = I_1 \Rightarrow \textcircled{1}$$

$$\frac{V_2 - V_1}{6} + \frac{V_2}{4} - I_2 = 0$$

$$-\frac{V_1}{6} + \frac{5V_2}{12} = I_2 \rightarrow \textcircled{2}$$

$$\frac{5V_2}{12} - I_2 = \frac{V_1}{6}$$

$$V_1 = \frac{5V_2}{2} - 6I_2 \rightarrow \textcircled{3}$$

$$\frac{2}{3} \left(\frac{5V_2}{2} - 6I_2 \right) - \frac{V_2}{6} = I_1$$

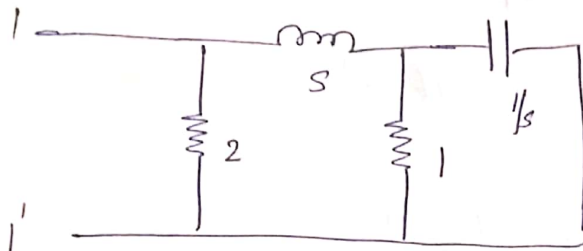
$$I_1 = \frac{5V_2}{3} - 4I_2 - \frac{V_2}{6}$$

$$I_1 = 1.5V_2 - 4I_2 \rightarrow \textcircled{4}$$

$$A = 2.5, B = 6, C = 1.5, D = 4 //$$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 2.5 & 6 \\ 1.5 & 4 \end{bmatrix}$$

(196)



$$Z_{11} = \left(1 \parallel \left(\frac{1}{s} + s \right) \right) \parallel 2$$

$$= \left(\frac{\frac{1}{s}}{1 + \frac{1}{s}} + s \right) \parallel 2$$

$$= \left(\frac{\frac{1}{s}}{\frac{s+1}{s}} + s \right) \parallel 2$$

$$= \left(\frac{1}{s+1} + s \right) \parallel 2$$

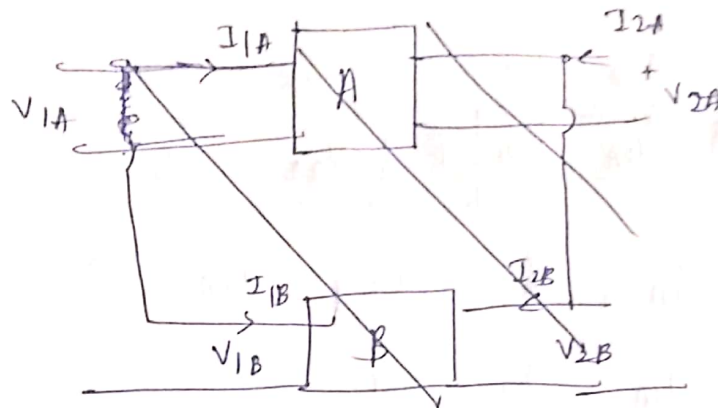
$$= \frac{s^2 + s + 1}{s+1} \parallel 2$$

$$= \frac{2(s^2 + s + 1)}{s+1}$$

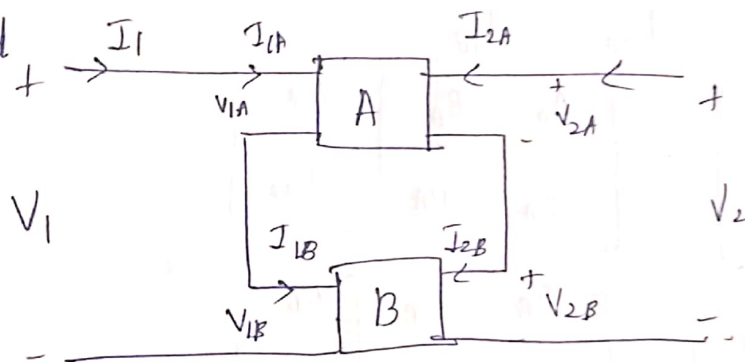
$$= \frac{2(s^2 + s + 1)}{\cancel{s+1} \frac{s^2 + s + 1 + 2s + 2}{s+1}}$$

$$= \frac{2(s^2 + s + 1)}{s^2 + 3s + 3}$$

20a



For series
connected
N/w



$$V_{1A} = Z_{11A} I_1 + Z_{12A} I_2 \rightarrow (1)$$

$$V_{2A} = Z_{21A} I_1 + Z_{22A} I_2 \rightarrow (2)$$

$$V_{1B} = Z_{11B} I_1 + Z_{12B} I_2 \rightarrow (3)$$

$$V_{2B} = Z_{21B} I_1 + Z_{22B} I_2 \rightarrow (4)$$

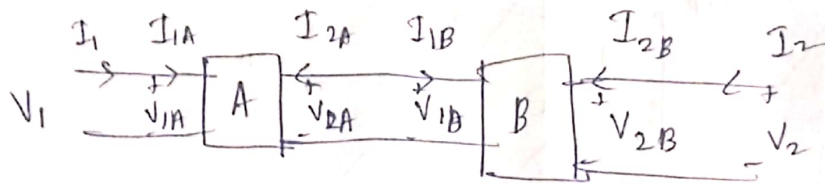
$$V_1 = V_{1A} + V_{1B}$$

$$= Z_{11A} I_1 + Z_{12A} I_2 + Z_{11B} I_1 + Z_{12B} I_2$$

$$V_1 = (Z_{11A} + Z_{11B}) I_1 + (Z_{12A} + Z_{12B}) I_2$$

$$V_2 = (Z_{21A} + Z_{21B}) I_1 + (Z_{22A} + Z_{22B}) I_2$$

For cascade connected n/w



$$V_1 = V_{1A}, \quad V_{2B} = V_2, \quad V_{2A} = V_{1B}$$

$$I_1 = I_{1A}, \quad I_2 = I_{2B}$$

$$I_{2A} = -I_{1B}$$

$$\begin{bmatrix} V_{1A} \\ I_{1A} \end{bmatrix} = \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} V_{2A} \\ -I_{2A} \end{bmatrix}$$

$$\begin{bmatrix} V_{1B} \\ I_{1B} \end{bmatrix} = \begin{bmatrix} A_B & B_B \\ C_B & D_B \end{bmatrix} \begin{bmatrix} V_{2B} \\ -I_{2B} \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} A_B & B_B \\ C_B & D_B \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

(Prob)

$$Y = \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

$$Z = Y^{-1} = \frac{1}{6-1} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \\ = \begin{bmatrix} 0.4 & 0.2 \\ 0.2 & 0.6 \end{bmatrix}$$

$$Z_A = Z_{11} - Z_{12} = 0.2 \Omega$$

$$Z_B = Z_{22} - Z_{12} = 0.4 \Omega$$

$$Z_C = Z_{12} = 0.2 \Omega$$

